Ryan Banerjee

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Education

University of California, Santa Barbara — Santa Barbara, CA
B.S. in Computer Engineering — *Expected June 2030*

- **Current Coursework:** Problem Solving with Computers I (C++ programming, computer architecture), Introduction to Electrical & Computer Engineering (circuit theory)

Projects

Theft-Deterrent Security System — July 2025 – Aug. 2025
Wharton Management & Technology Summer Institute, UPenn – Program GPA: 4.0 — *Philadelphia, PA*

- Led a **team of 5** to build a **Raspberry Pi** device that protects personal belongings from theft
- Used a camera with **OpenCV** to detect when a person approaches the belongings and whether an item is taken
- On detection, sends an **alert to the owner's phone** and triggers an **alarm and flashing light**; live camera feed viewable from the phone at any time
- Directed the team's CAD design in **SolidWorks**; built and programmed **Arduino circuits in C++** in program labs
- Validated demand through **online surveys and interviews**, then pitched the product as a startup venture

AI Crash-Detection Helmet — July 2024 – Aug. 2024
Berkeley Business Academy for Youth, UC Berkeley Haas – 1 of 50 selected — *Berkeley, CA*

- Conceived a helmet that uses **AI to detect crashes** and sends parents a **GPS location alert**; produced design drawings
- Validated the concept through **surveys and interviews**; pitched to **venture capitalists** and Berkeley professors

Skills

Languages: C++, Python, Shell scripting
Hardware: Raspberry Pi, Arduino, oscilloscope, multimeter
Software & ML: OpenCV, AWS (SageMaker, S3, EC2), machine learning
CAD & Fabrication: SolidWorks, OnShape, machine shop fabrication
Business: Entrepreneurship, pitch decks, market research, fundraising, public speaking

Experience

FIRST Robotics Competition Team 7419 — Sep. 2022 – June 2026
Hardware & Business Member → Business Lead → Mentor — *Dublin, CA*

- Designed the robot's **end effector** in **OnShape** and fabricated it using machine shop equipment
- Led the **15-member** Business department on a **100+ member** team, overseeing **1,000+ hours** of community outreach
- Raised **$10K+** in grants and sponsorships, including a **NASA sponsorship**
- Won the **FIRST Engineering Inspiration Award** through the Business department's award submission, qualifying the team for the **2024 FIRST World Championship**
- Mentored K–8 students in robotics and led STEM outreach in **India**

Amazon Web Services (AWS) — Mar. 2025 – Apr. 2026
Student Intern, Machine Learning — *Remote*

- Built an ML model that detects diabetes with **84%+ accuracy** using **SageMaker, S3, and EC2**
- Team was **1 of 6 (out of 12)** selected to present results to Amazon executives

Cisco — July 2026
High School Extern — *New York City, NY*

- Built a pitch deck mapping Cisco products and services to Taco Bell's **AI voice-ordering drive-thrus** and **inventory tracking**, projecting **$15M+ in annual savings**
- Presented to industry professionals and **placed 2nd** among competing teams

Interests

Badminton, community service, embedded systems, VLSI design, computer architecture